

week-5(Pravalika-2520030402)

Question:

Implement a producer-consumer communication system using anonymous pipes where the parent produces data and the child consumes it.

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    int fd[2];
    pid_t pid;

    printf("PRODUCER-CONSUMER USING PIPE\n");

    pipe(fd);

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        char buffer[100];

        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Child (Consumer): %s\n", buffer);

        close(fd[0]);
    }
}
```

```

        exit(0);
    }
    else
    {
        char message[] = "Data sent by Parent (Producer)";

        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        printf("Parent (Producer): Data sent\n");

        close(fd[1]);

        wait(NULL);
    }

    printf("\nExecuting: ls -l | grep \".c\"\n");

    int pipefd[2];
    pipe(pipefd);

    pid_t p1 = fork();

    if (p1 == 0)
    {
        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("exec ls");
        exit(1);
    }

    pid_t p2 = fork();

    if (p2 == 0)
    {
        close(pipefd[1]);

        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);
    }

```

```
    execlp("grep", "grep", ".c", NULL);

    perror("exec grep");
    exit(1);
}

close(pipefd[0]);
close(pipefd[1]);

wait(NULL);
wait(NULL);

return 0;
}
```

Output:

```

rikki@BOOK-GJU4MP9458:~$ nano practical5.c
rikki@BOOK-GJU4MP9458:~$ gcc practical5.c
rikki@BOOK-GJU4MP9458:~$ ./a.out
PRODUCER-CONSUMER USING PIPE
Parent (Producer): Data sent
Child (Consumer): Data sent by Parent (Producer)

Executing: ls -l | grep ".c"
drwxr-xr-x 3 rikki rikki 4096 Aug  3 04:34 2520030002_Practical
drwxr-xr-x 3 rikki rikki 4096 Aug  3 04:34 2520030003_Practical
drwxr-xr-x 4 rikki rikki 4096 Aug  3 04:35 2520030402_Practical
-rw-r--r-- 1 rikki rikki 146 Jul 28 05:03 addition.c
-rw----- 1 rikki rikki 147 Jul 28 04:56 addition.c.save
-rw-r--r-- 1 rikki rikki 140 Aug  3 06:57 execsc.c
-rw-r--r-- 1 rikki rikki 187 Aug  3 07:05 execsc1.c
-rw-r--r-- 1 rikki rikki 204 Aug  3 08:07 execsc2.c
-rw-r--r-- 1 rikki rikki 218 Aug  3 08:13 execsc3.c
-rw-r--r-- 1 rikki rikki 237 Aug  3 08:20 execsc4.c
-rw-r--r-- 1 rikki rikki 471 Aug  3 08:37 execsc5.c
-rw-r--r-- 1 rikki rikki 562 Aug 11 05:36 expl.c
-rw-r--r-- 1 rikki rikki 779 Aug  4 05:23 exp2.c
-rw-r--r-- 1 rikki rikki 242 Aug  4 05:58 filesystem.c
-rw-r--r-- 1 rikki rikki 146 Jul 28 04:59 gcc addition.c -o add
-rw-r--r-- 1 rikki rikki 436 Aug 25 05:49 open.c
-rw-r--r-- 1 rikki rikki 183 Aug  1 09:29 p2.c
-rw-r--r-- 1 rikki rikki 269 Aug  1 09:44 p3.c
-rw-r--r-- 1 rikki rikki 282 Aug  1 09:55 p4.c
-rw-r--r-- 1 rikki rikki 210 Aug  1 10:11 p5.c
-rwxr-xr-x 1 rikki rikki 16352 Aug 22 09:53 practical1
-rw-r--r-- 1 rikki rikki 908 Aug 22 09:51 practical1.c
-rw-r--r-- 1 rikki rikki 728 Aug 22 10:04 practical2.c
-rw-r--r-- 1 rikki rikki 729 Sep 12 09:02 practical3.c
-rw-r--r-- 1 rikki rikki 813 Sep 12 09:28 practical4.c
-rw-r--r-- 1 rikki rikki 1423 Sep 12 09:35 practical5.c
-rw-r--r-- 1 rikki rikki 687 Aug 22 10:13 skill1.c
-rw-r--r-- 1 rikki rikki 827 Aug 22 10:19 skill2.c
-rwxr-xr-x 1 rikki rikki 16096 Jul 28 05:25 switch
-rw-r--r-- 1 rikki rikki 442 Jul 28 05:25 switch.c
rikki@BOOK-GJU4MP9458:~$ |

```